Dino Claro

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Profile

Master's student in Robotics at ETH Zürich with an interest in bio-inspired robotics. Focused on translating nature's engineering principles into robotic intelligence, drawing inspiration from a love of escaping into nature.

Education

MSc Robotics, Systems and Control — 2025 – Expected Mar. 2027
ETH Zürich, Switzerland
Semester Project: *Offline Model-Based Reinforcement Learning for a Electrohydraulic Musculoskeletal Robot*
Relevant Coursework: Robot Control and Modelling, Machine Learning for Robotics, Computer Vision.

BSc (Eng) Mechanical & Mechatronic Engineering — 2020 – 2023
University of Cape Town (UCT), South Africa — GPA: 91.76 | Faculty Rank: 1st
Thesis Title: *Object Detection for the Duckiebot* (link)
Relevant Coursework: Design for Manufacturing, Mechanics & Dynamics, Control Systems, Embedded C

National Senior Certificate (High School) — 2015 – 2019
King Edward VII School, South Africa — Average: 89.0 | Grade Rank: 3rd

Experience

African Robotics (ARU): *Research Internship* (link) — Jan. – Aug. 2025, *Cape Town*
- ARU is a research lab at UCT, seeking to solve robotics challenges through a uniquely African perspective.
- Designed and manufactured a 3-DOF actuated spine for the realisation of a robot-physical model of the cheetah's hindquarters by attaching the 3-DOF spine joint to an actuated tail and bipedal robot known as Baleka.

MechatronicSystems.Group: *Vacation Work* — Jul. 2023, *Cape Town*
- Research unit within UCT, focusing on control systems, system identification, and embedded hardware.
- Investigated the potential of the Duckiebot robot as a teaching and project platform for undergraduate students by implementing simple locomotion control policies.

S-Plane: *Embedded Systems Internship* — Dec. 2022, *Cape Town*
- S-Plane specialises in delivering automation solutions for aircraft systems.
- Developed a synthesizable VHDL model of a popular ADC and a Python package for parsing C++ code.

Related Projects

Offline Model-based RL for an Electrohydraulic Musculoskeletal Robot: *Semester Project* — *Currently*
- Fitted robot leg with joint sensors and developed a data collection pipeline to capture motion trajectories.
- Collected trajectories are used to train an offline learned dynamics model which allows an RL agent to learn a control policy for the robot.

ORCA Dexterous Hand Modification (link) — Sep. – Dec. 2025
- Engineered a biomechanically-inspired modification for the open-source ORCA hand, introducing a coupled PIP-DIP with the goal of improving grasp on small-diameter objects.
- Expanded the fingertip workspace and increased total DOFs to 21 without adding any additional actuators by using a bio-inspired, coupled tendon design.

Learning Dexterous Manipulation: Real World Robotics ETH HS 2025 (link) Sep. – Dec. 2025

- Trained and deployed a Reinforcement Learning policy for an in-hand manipulation task, achieving 6.0 rotations/min of a cube.
- Developed Imitation Learning policies for object colour-sorting, using DINOv3 as a visual backbone and a Diffusion Policy.

Monocular Visual Odometry Pipeline (Tokyo Drift) (link) Sep. – Dec. 2025

- Implemented a monocular 6-DOF camera pose estimation pipeline from video streams, tuned for common autonomous driving datasets (KITTI, Malaga). The pipeline produced locally accurate trajectories across all datasets but suffered from scale drift, earning it the affectionate name *Tokyo Drift*.
- Current efforts focus on integrating IMU measurements to mitigate scale drift.

Skills & Interests

Technical Skills	Design for manufacturing, rapid prototyping, robotic control and learning
Programming & Software	SolidWorks, MATLAB, Python, ROS, C
Technical Interests	Soft robotics design and simulation, bio-inspired robotics, robotic perception
Languages	English (native), Portuguese (intermediate), Afrikaans (intermediate)
Personal Interests	Mountaineering, hiking, camping, cooking

Leadership & Activities

UCT Mechanical Engineering Department TA (2022-2024): Mechanical Engineering Department tutor for second-year Solid Mechanics, third-year Control Systems, and fourth-year Mechanical Vibrations.

UCT Residence Committee IT Representative (2021): Managed minor issues with the residence's computers and Wi-Fi, and was responsible for reporting technical issues for resolution.

Scholarships & Awards

Detailed descriptions of all scholarships and awards are available on my personal portfolio.

- **Skye Foundation Postgraduate Scholarship (2025):** Awarded to South African students who demonstrate exceptional academic merit to pursue research at the University of South Africa
- **ECSA Medal of Merit (2023):** Top 4th-year Engineering Student
- **Top 10 Final-Year Projects (2023):** leading institutions ; valued at CHF 28,000 for studies at ETH Zürich.
- **Klaus-Jürgen Bathe Scholarship (2023):** Awarded to students in the final two years of undergraduate study who show evidence of high intellectual ability and commitment to achieving excellence in engineering. Each award is valued at R35,000.
- **EBE Faculty Scholarship (2022, 2023):** highest GPA within the UCT Engineering & Built Environment Faculty for the respective year

Short Courses

HarvardXCS50 (2024): Introduction to computer science covering foundational programming concepts, algorithms, and problem-solving using C, Python, and web development.

Dale Carnegie – Generation.Next (2019): World-renowned course aimed at developing leadership and communication skills in high school students.

Gordon Institute of Business Science – Spirit of the Youth (2018): Provides high-impact learners with a platform to articulate their vision for South Africa, engaging in dialogues with policymakers and peers on crucial issues.